

ADVANCE INFORMATION

PAPER 1 4055/1 THEORY, DRAWING AND DESIGN

Each candidate will require drawing instruments and **one** piece of good quality **A2** drawing paper. This should be ruled out before the examination.

PAPER 2

INSTRUCTIONS TO SUPERVISORS

Please note that all the tools, materials and equipment required for this test are to be provided by the centre.

They will not be provided by the Zimbabwe School Examination Council.

In addition to drilling machines, candidates are encouraged to use available machines in the workshop such as milling machines, centre lathe, shaping machines, bending machines as well as grinding machines.

NB: The reduction of material to examination requirements and the preparation of special tools or parts must be undertaken by the candidates before the examination. Failure to prepare materials as specified will seriously handicap candidates.

Each candidate should be supplied with the stated materials.

If Bright Drawn mild steel is not available, Black Mild Steel (BMS) will be acceptable but the surfaces and edges should be cleaned.

CONFIDENTIAL INFORMATION TO THE SUPERVISOR

Candidates are expected to have covered the following skills.

- 1 Ability to interpret the working diagrams.
- 2 Measuring and marking out using various tools.
- 3 Cutting using various tools.
- 4 Drilling
- 5 Riveting
- 6 Shaping especially using a file.
- 7 Formation of slot and tang.
- 8 Bending
- 9 Thread cutting
- 10 Brazing/Silver soldering
- 11 Force fitting
- 12 Finish by draw filing/machining.
- 13 Ability to join different parts together to build an artefact (assembling).

THE FOLLOWING MATERIALS ARE REQUIRED

PART 1

One piece **100 mm × 30 mm × 3 mm** BDMS/cleaned MS both ends squared.

PART 2

One piece **40 mm × 30 mm × 3 mm** BDMS/cleaned MS both ends squared.

PART 3

One piece **35 mm × 30 mm × 10 mm** BDMS/cleaned MS both ends squared.

PART 4

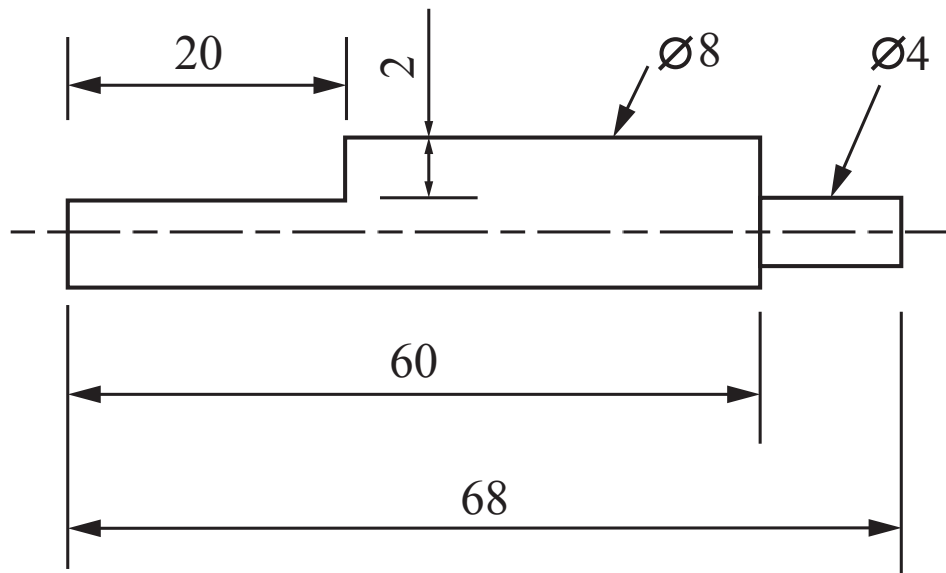
One piece **50 mm × 20 mm × 3 mm** BDMS/cleaned MS both ends squared.

PART 5

One piece **30 mm × Ø6 mm** BDMS/cleaned MS both ends squared (prepared before the examination).

PART 6

One piece 68 mm × Ø8 mm BDMS/cleaned BMS (prepared before the examination).



One strong plastic bag in which to enclose work.

Two-tie-on-labels and a string.

TOOLS AND EQUIPMENT REQUIRED

300 MM flat files
Warding files
Needle file/three squares
300 mm steel rules
Scribers
M6 Tap and wrenches
M6 Die nut and stock
Silver solder/brazing sticks
Brazing flux
Flat chisels
Ball pain hammers
Hacksaws
Centre ouches
Pair of dividers
Countersunk bits
Folding bars
Oxy-acetylene gas set/handigas set
Pillar drill machine/hand drill
Ø2 drill bits (hss)
Ø3drill bits (hss)
Ø5 drill bits (hss)
Ø4 drill bits (hss)
Ø8 drill bits (hss)
Clear goggles
Flat pliers
Coolant
Brazing goggles